Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lessons 1–2 — Triangle Inequality Theorem; Converse of the Pythagorean Theorem	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.3

2. Standards

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Must. Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

Must not. Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: three lengths as sticks and as one inequality
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 3, 5 and 7 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	board 7: a reason checked against the squares, not the sum

4. Learning Target and Essential Question

Learning target. I can decide whether three lengths can make a triangle, and whether they make a right triangle.

Essential question. Why can't any three sticks make a triangle, and how do three numbers tell you a corner is square?

Building on. Grade 7 (MA.7.NSO.1.1) — squares of whole numbers; 4.05 — square roots of perfect squares; Grade 5 (MA.5.GR.1.1) — a right triangle has one right angle.

Working toward. 6.03 — the Pythagorean Theorem finds a missing side; 6.05–06 — distance on the coordinate plane; Algebra 1 and Geometry — the Triangle Inequality in proofs.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two sides of a triangle are 5 and 9. Why can the third side not be 4, and why can it not be 14?	DOK 3
Lengths 6, 8 and 10 make a right triangle. Without computing, do 12, 16 and 20? Why?	DOK 2
A student tests $13^2 + 12^2 = 5^2$ for sides 13, 12 and 5 and decides it is not a right triangle. What should she have tested?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Can sides of 2 ft, 7 ft and 10 ft make a triangle?	***No — 2 + 7 = 9 is less than 10.***	Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners]
2	Can sides of 6 m, 6 m and 11 m make a triangle?	***Yes — 6 + 6 = 12 is more than 11.***	No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle]
3	Which three lengths can make a triangle?	***B — 5, 11, 8***	(A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) 4 + 6 = 10 is less than 12: those sides cannot meet [Florida B1G-M, GR.1.3: believing the sides need not meet at the vertices]
4	Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side.	***5 < x < 19 — more than 5 cm and less than 19 cm.***	5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach]
5	Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side?	***C — 4 < x < 16***	(A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference]
6	Do sides of 8 inches, 15 inches and 17 inches make a right triangle?	***Yes — $8^2 + 15^2 = 289 = 17^2$.***	No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]

#	What the board asks	Answer	If the board is wrong
7	A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason?	***A — yes, because $7^2 + 24^2 = 25^2$***	(B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III]
8	A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle?	***Yes — $20^2 + 21^2 = 841 = 29^2$.***	'Cannot tell without the angles' [Notes III: the sides are enough]
9	Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain.	***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.***	'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines stay on the board all period: "the two SHORTER sides, added, beat the longest" and "the two shorter sides' squares, added, EQUAL the longest side's square". Every board starts by circling the longest side. Popsicle-stick sets for Notes I if a picture is not enough.

ELL. **Inequality** means "not equal" — here, MORE than. **Hypotenuse**: the long side across from the square corner. Say "longest side" with every board, and point to it.

Enrichment. Board 4 finishes early for some: how many whole-number lengths can the third side have when the other two are 7 and 12? (Thirteen: 6 through 18.) Then: which of them make a right triangle? (None is a whole number: √95 and √193.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Ana's test (6 + 7 > 9) proves only that the three sides make a triangle; the converse is a test of squares, and 85 is not 81. A board that says 'she is wrong' without the squares has the verdict and not the reason.

Vocabulary. **triangle inequality**; **legs**; **hypotenuse**; **converse**

Materials. Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Triangle inequality — 2N5; Converse of the Pythagorean theorem: is it a right triangle? — EQZ.

Pythagorean Theorem

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 3 — Pythagorean Theorem	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.1

2. Standards

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must. Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

Must not. Stories (6.04); the coordinate plane (6.05); radicands over 225; simplifying a radical ($\sqrt{50} = 5\sqrt{2}$ is Algebra 1 — a root's decimal is the answer); 3-D.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes III and boards 3–4: every leg is checked against c — a leg is always shorter than the hypotenuse
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I and board 5: one theorem however the triangle is turned — the square corner first

4. Learning Target and Essential Question

Learning target. I can find a missing side of a right triangle — the hypotenuse, or a leg.

Essential question. If you know two sides of a right triangle, how do you find the third — and how do you know which side is c?

Building on. 6.01–02 — the converse and the 3-4-5 squares; 4.05 — square roots of perfect squares; Unit 2 (MA.8.NSO.1.2) — a square root between two whole numbers.

Working toward. 6.04 — the theorem inside real situations; 6.05–06 — distance on the coordinate plane; Geometry — right-triangle trigonometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A right triangle has legs 5 and 12. Why is the hypotenuse 13 and not 17?	DOK 2
Without computing, which is longer: the hypotenuse when the legs are 6 and 8, or a leg when the hypotenuse is 10 and the other leg is 6? How do you know?	DOK 3
Why can a missing leg never be found by adding two squares?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse?	***15 cm***	21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]
2	A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth?	***$\sqrt{97}$ in ≈ 9.8 in***	13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice, 9.849 → 9.85 → 9.9: round once, from the full value
3	A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg?	***15 ft***	32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]
4	A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg?	***A — 9 m***	(B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract]
5	Two sides of this right triangle are 5 cm and 7 cm. How long is the third side, to the nearest tenth?	***$\sqrt{74}$ cm ≈ 8.6 cm***	4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]
6	A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x?	***D: $16^2 + x^2 = 20^2$, so x = 12 in***	(A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) squared the legs but not the hypotenuse — the 20 is a side too: 20^2 [Florida B1G-M, GR.1.1: calculation errors with the theorem]
7	A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth?	***$\sqrt{115}$ m ≈ 10.7 m***	16.6 — added the squares, longer than the hypotenuse [Notes III]
8	A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth?	***$\sqrt{185}$ in ≈ 13.6 in***	19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I]
9	Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg?	***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.***	'13.9 is fine' — no check against the hypotenuse [Notes III]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. $\text{Leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$ stays on the board all period. Every board starts by marking the square corner and the side across from it. A set-up with the numbers already squared for boards 1–2 if the squaring is the obstacle.

ELL. **Hypotenuse**: the long side across from the square corner. **Leg**: one of the two sides that make the square corner. Say "square it" and "square root" every time, with the gesture of a square.

Enrichment. Legs 1 and 1 give $\sqrt{2}$; legs $\sqrt{2}$ and 1 give $\sqrt{3}$ — keep going (the Wheel of Theodorus): which leg lengths give $\sqrt{7}$? And why is every hypotenuse in the wheel $\sqrt{n}$?

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Jo added the squares to find a leg, and her answer is longer than the 13-cm hypotenuse — impossible. A board that finds 12 without saying why 13.9 cannot be a leg has the number and not the check.

Vocabulary. **Pythagorean Theorem**; **hypotenuse**; **square root**

Materials. Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: find the missing leg or hypotenuse length — MTM.

Pythagorean Theorem in Real-World Problems

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 4 — Pythagorean Theorem in Real-World Problems	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.1

2. Standards

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must. Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

Must not. 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	every board is a real object: a pool, a ladder, a ramp, a lawn, a drone — the triangle is found in it before any number
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	boards 7 and 9: the answer is checked against the story — does it fit, can it reach

4. Learning Target and Essential Question

Learning target. I can find a length in a real situation by finding the right triangle inside it.

Essential question. Where is the right angle in a ladder leaning on a wall — and why does that decide which length is c?

Building on. 6.03 — a missing hypotenuse or leg; 5.04 — real lengths from a plan; Grade 3–4 — a rectangle's corners are right angles.

Working toward. 6.05–06 — the same right triangle drawn on a grid between two points; Geometry — right-triangle trigonometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A ladder's foot slides farther from the wall. Does the top go up or down? Why?	DOK 3
Why is the diagonal of a rectangle always longer than either of its sides?	DOK 2
A 10-ft ladder reaches 9.5 ft up a wall. Is its foot more or less than 3 ft from the wall? How can you tell without the theorem?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A swimming pool is 12 m long and 5 m wide. How long is its diagonal?	***13 m***	17 — added the sides [MA.8.GR.1.1: the theorem is about squares]; 10.9 — subtracted, 144 − 25, taking the 12 as c [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]
2	A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach?	***12 ft***	17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]
3	A ramp rises 4 ft over a run of 11 ft. How long is the ramp, to the nearest tenth of a foot?	***About 11.7 ft***	15 — added the rise and the run [MA.8.GR.1.1]; '11, the same as the run' — the ramp is the hypotenuse, longer [Notes I]
4	A 26-ft ladder reaches 24 ft up a wall. How far from the wall is its foot?	***A — 10 ft***	(B) added the squares: about 35.4 ft, farther than the ladder is long [Notes III: for a leg, subtract]; (C) subtracted the lengths, 26 − 24 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 100$ and forgot the square root [Math Nation names skipping the final root]
5	A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk?	***6 m shorter***	15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk
6	A drone is 30 m from Maya, straight above a spot on the ground 18 m from her. Which length is the hypotenuse?	***C — the 30 m to the drone***	(A) took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the drone [Florida B1G-M, GR.1.1: failing to visualize the triangle in a real-world problem]
7	A box is 7 inches wide and 12 inches long. Will a 14-in umbrella lie flat in it, corner to corner?	***No — the diagonal is about 13.9 in.***	'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 193 > 14' — forgot the root [Math Nation names skipping the final root]
8	A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot?	***About 7.1 ft***	12.2 — added the squares (√149): higher than the ladder is long [Notes III]
9	A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands?	***The height is a leg — always shorter than the 11-ft ladder.***	'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every story starts with a sketch: the right angle marked first, then c. Boards 1–3 come with the sketch printed; the student labels it. $Leg^2 + leg^2 = hypotenuse^2$ stays on the board.

ELL. **Ladder**, **wall**, **ramp**, **diagonal**, **kite string**: point at the picture for each before the board. "How high" is a leg; "how long is the ladder" is the hypotenuse.

Enrichment. A 25-ft ladder stands 7 ft from a wall. Its foot slides 8 ft farther out. How far does the top slide down? (24 − 20 = 4 ft — both heights whole.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the ladder is the hypotenuse, the longest side, and the height it reaches is a leg — so an 11-ft ladder can never touch a 12-ft window, wherever its foot stands. A board that tries numbers has not named the hypotenuse.

Vocabulary. **diagonal**; **rise and run**

Materials. Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: word problems — 87U.

Distance on the Coordinate Plane

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 5 — Distance on the Coordinate Plane	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.2

2. Standards

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must. Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

Must not. The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 3 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes I and board 9: a distance across zero checked by counting the squares

4. Learning Target and Essential Question

Learning target. I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

Essential question. How can two pairs of numbers tell you how far apart two points are?

Building on. 6.03–04 — the Pythagorean Theorem for a missing side; Grade 6 (MA.6.GR.1.2) — the distance between points with the same x or the same y; Grade 7 — subtracting integers.

Working toward. 6.06 — perimeters and maps on the grid; Algebra 1 and Geometry — the distance formula, which is this triangle written as one line.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is the distance from (−4, 3) to (5, 3) not 5 − 4?	DOK 2
Two points have different x's and different y's. Why can you not just subtract?	DOK 3
Does it matter which point you start from? Why does the answer come out the same?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	How far apart are $(-7, -4)$ and $(-7, 6)$?	***10 units***	2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]
2	How far apart are $(-9, 2)$ and $(3, 2)$?	***12 units***	6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]
3	What is the distance between (−5, 1) and (3, 1)?	***B — 8 units***	(A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Florida B1G-M, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I]
4	How far apart are $A(-4, -3)$ and $B(5, 9)$?	***15 units***	21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn']
5	How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth?	***$\sqrt{89} \approx 9.4$ units***	13 — added the legs [MA.8.GR.1.2]; $\sqrt{(1^2 + 8^2)}$ — subtracted 3 − 2 across zero [Florida B1G-M, GR.1.2]
6	Which gives the distance between (4, 5) and (−3, −1)?	***C — $\sqrt{(7^2 + 6^2)} = \sqrt{85}$***	(A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum]
7	How far is $(5, -12)$ from the origin?	***13 units***	17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares]
8	On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$. How far is the school from the library?	***7 mi***	1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]; a distance to the town hall — the wrong pair of points
9	Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance?	***He subtracted 7 − 3. Across zero the pieces add: 10 units.***	'4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A printed grid for every board. The dashed legs are drawn on Notes II–III and the examples; from board 4 the student draws them. "Across, then up or down" said aloud with every slanted board.

ELL. **Horizontal**: side to side, like the horizon. **Vertical**: up and down. **Distance**: how far apart — never negative. Point at the grid with each word.

Enrichment. Find every point on the grid exactly 5 units from the origin with whole-number coordinates. (There are 12: (±5, 0), (0, ±5), (±3, ±4), (±4, ±3).)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Ty subtracted 7 − 3, but −3 and 7 are on opposite sides of zero, so the two pieces add: 3 + 7 = 10. A board that says only '10' without naming Ty's step has the answer and not the error.

Vocabulary. **horizontal distance**; **vertical distance**; **legs on the grid**

Materials. Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Find the distance between two points on a horizontal or vertical line — L9V; Find the distance between two points — ZBP.

Real-World Coordinate-Plane Problems; Perimeter

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 6 — Real-World Coordinate-Plane Problems; Perimeter	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.2

2. Standards

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must. Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

Must not. The distance formula (Clarification 2); radicands over 225; simplifying or combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes II and boards 5, 7 and 8: maps, the streets against the straight line

4. Learning Target and Essential Question

Learning target. I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

Essential question. How do you measure around a shape whose sides do not follow the grid lines?

Building on. 6.05 — the distance between two points on the grid; 6.03 — the Pythagorean Theorem; Grade 6 (MA.6.GR.1.3) — the perimeter of a rectangle on the coordinate plane.

Working toward. 6.07–08 — similar triangles; Geometry — perimeters on the plane, and the distance formula, which is this triangle written as one line.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is the straight path shorter than the streets unless the two points share an x or a y?	DOK 2
Notes III rounded once, at the end. When does rounding each side first change the answer?	DOK 3
A triangle has no right angle. Where would you draw a line to give it two?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Triangle ABC has corners $A(-6,-2)$, $B(6,-2)$ and $C(6,3)$. Find its perimeter.	***30 units***	17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum]
2	A triangle has corners $(-3,-4)$, $(1,-4)$ and $(1,5)$. Find its perimeter, to the nearest tenth.	***22.8 units***	13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2]
3	Triangle $A(-4,-6)$, $B(10,-6)$, $C(1,6)$ has no right angle. Find its perimeter.	***42 units***	41.8 — halved the base, 7 and 7, as Notes III's triangle allowed: 14 + 2√193 [the foot is at (1, −6): 5 and 9]; 40 — copied AC's 13 onto BC; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle]
4	What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth?	***D — 19.1 units***	(A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors]
5	On a map in blocks, the library is at $(-7,-1)$ and the pool at $(5,4)$. How much shorter is the straight line than the streets?	***4 blocks shorter***	13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis]
6	How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)?	***A — 15 units***	(B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths]
7	On a park map in meters, the fountain is at $(-5,2)$ and the gate at $(4,-3)$. How far is it straight across, to the nearest tenth?	***√106 ≈ 10.3 m***	14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis]
8	On a trail map in miles, the camp is at $(-3,-6)$, the lake at $(2,6)$ and the trailhead at $(-1,0)$. How far is the camp from the lake, in a straight line?	***13 mi***	17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points

#	What the board asks	Answer	If the board is wrong
9	Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth.	***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.***	'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A printed grid for every board. Find the sides on grid lines first and write them on the figure; then the slanted side, by its triangle. "Every side counts" said aloud with every perimeter.

ELL. **Perimeter**: around the outside — every side. **Along the streets**: on the grid lines. **Straight line**: one segment, as a bird flies. Trace each with a finger on the grid.

Enrichment. Plot (0, 0), (6, 0) and (3, 4). Is it a right triangle? (No: 25 + 25 is not 36.) Find its perimeter. (5 + 5 + 6 = 16.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Mia counted the squares her segment crosses, but a slanted segment is the hypotenuse of a right triangle — legs 3 and 1, √10 ≈ 3.2, longer than 3. A board that says only '3.2' has the length and not the reason.

Vocabulary. **perimeter**; **height of a triangle**; **straight-line distance**

Materials. Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: find the perimeter — VGE.

Similar Triangles; Indirect Measurement

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lessons 7–8 — Similar Triangles; Indirect Measurement	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.2.4

2. Standards

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Must. Decide similarity from all three side ratios; match the sides of a turned pair (by size, or by a name's letters); find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

Must not. AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: one pair as a picture, three ratios and one k
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 5 split; the re-vote
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes III, Example 2 and boards 6–8: heights measured without climbing

4. Learning Target and Essential Question

Learning target. I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

Essential question. How can a shadow tell you the height of something you cannot climb?

Building on. 5.01 — the scale factor k = copy ÷ original; Grade 7 (MA.7.GR.1.5) — scale drawings; 6.03 — right triangles.

Working toward. The Unit 6 test tomorrow; Unit 10 — slope, where similar triangles on one line give one rise over run; Geometry — similarity proved (AA, SAS, SSS).

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Every side of a triangle is doubled. What happens to its angles, and to its shape?	DOK 2
Adding 2 to every side of a 3, 4, 5 triangle makes a triangle that is not similar. Is there a triangle where adding the same amount to every side does make a similar one? (Yes: an equilateral triangle — its sides stay equal to each other.)	DOK 3
The shadow method needs both shadows measured at the same time. What would go wrong an hour apart?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?	***Yes — every ratio is 1.5***	'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three]
2	Triangle ABC is similar to triangle DEF. Find x.	***x = 24***	13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 20 — added 12 to the 8 [Notes I: similar means multiplied by one k]
3	A triangle has sides of 12, 16 and 20 inches. A similar triangle's longest side is 5 inches. Find its other two sides.	***3 inches and 4 inches***	48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]
4	The two triangles are similar (the second is turned). Which side of the second triangle matches the 9?	***C — the 12***	(A) the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) looked for a side equal to the 9 — similar taken as equal [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]
5	Which triangle is similar to one with sides 5, 6 and 8?	***B — 10, 12, 16***	(A) added 2 to every side [Notes I: similar means multiplied by one k, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]
6	A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree?	***24 ft***	about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied]
7	A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A?	***2.4 m***	1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied]
8	Lena's eyes are 5.5 ft up. In a 10-inch mirror on the ground 6 ft away, she sees the top of a flagpole 24 ft past the mirror. The two right triangles are similar. How tall is the flagpole?	***22 ft***	about 1.4 — divided the other way, 5.5 × 6 ÷ 24 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 13.2 — used the 10 inches for the 6 ft, 24 ÷ 10 × 5.5 [the extra number]
9	Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain.	***No — similar means × one k, not + 2.***	'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A three-row table on every board — shortest, middle, longest — with the original's sides, the copy's sides and copy ÷ original. Sort both triangles' sides before matching anything.

ELL. **Similar**: the same shape, any size. **Corresponding**: matching. **Scale factor**: the number every side is multiplied by. Point to each matching pair as it is said.

Enrichment. Notes III's tree is 30 ft. A cell tower's shadow at the same moment is 80 ft: how tall is the tower? (120 ft.) Is that realistic? (Yes: most cell towers are 50–200 ft — the state guide's own question.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Kim added 2 to every side, but similar sides are MULTIPLIED by one k — 5/3, 6/4 and 7/5 are three different numbers. A board that says only 'not similar' has the verdict and not the reason.

Vocabulary. **similar triangles**; **corresponding sides**; **scale factor k**; **indirect measurement**

Materials. Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Side lengths of similar triangles — QSK; Similar triangles and indirect measurement — 88W; Identify similar triangles — 6PD.

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Thread B · Day 1 — Angle Pairs: Supplementary, Complementary, Vertical, Adjacent	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.4

2. Standards

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

Must. Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

Must not. Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II–III: one pair as a picture, a name and an equation
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 7 split; the re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes III and boards 5–6: the same equation shape behind every pair

4. Learning Target and Essential Question

Learning target. I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

Essential question. When two angles are neighbours or opposites, what does one tell you about the other?

Building on. Grade 4 (MA.4.GR.1.3) — angle measures add, a straight angle is 180°; Unit 1 — two-step equations (MA.7.AR.2.2). An x on both sides is new: Example 2 takes it off one side as its own line.

Working toward. T-B2 tomorrow — the angles of a triangle; Geometry — angle pairs made by parallel lines and a transversal.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 11 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Can two adjacent angles be vertical angles? Why or why not?	DOK 2
Two angles are supplementary and equal. What are they, and what do they make?	DOK 2
Board 6's x is 12 and its angles are 52°. Which does the question ask for, and how do you know?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	What is the complement of 34°?	***56°***	146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]
2	What is the supplement of 112°?	***68°***	−22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]
3	Two lines cross. One angle is 47°. Find a and b.	***a = 47°, b = 133°***	a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]
4	$\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28°$. What is $m\angle Q$?	***B — 62°***	(A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add to 90]
5	A linear pair measures $(3x+15)°$ and $105°$. Find x.	***x = 20***	30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4]
6	Vertical angles measure $(6x-20)°$ and $(2x+28)°$. Find the measure of each angle.	***52°***	12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary]
7	In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent?	***C — angles AOB and BOC***	(A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side]

#	What the board asks	Answer	If the board is wrong
8	Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x.	***x = 142***	92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4]
9	Can two vertical angles be complementary? Always, sometimes or never? Explain.	***Sometimes — only when each is 45°.***	'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. One card on the desk: Corner (90) — complementary; Straight (180) — supplementary; across — vertical, equal; next to, sharing a side — adjacent. Name the pair before writing a number.

ELL. **Adjacent**: next to. **Vertical** here means across from (not up and down). **Complement / supplement**: the piece that completes 90 / 180. Point and say each one.

Enrichment. Three lines cross at one point and make six angles. Two of them are 40° and 75°. Find the other four. (40°, 75°, 65°, 65°: vertical pairs are equal, and the three around one side make 180.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: vertical angles are equal, so they are complementary only when each is 45° — sometimes, never always. A board that says 'yes' or 'no' without the 45 has the verdict and not the reason.

Vocabulary. **adjacent angles**; **vertical angles**; **complementary angles**; **supplementary angles**

Materials. Whiteboards and markers. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 11 test: Identify complementary, supplementary, vertical, adjacent, and congruent angles — HGV; Find measures of complementary, supplementary, vertical, and adjacent angles — R2B; Write and solve equations using angle relationships — DBV.

Interior and Exterior Angles of a Triangle

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Thread B · Day 2 — Interior and Exterior Angles of a Triangle	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.5

2. Standards

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Must. Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

Must not. Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I–II: 180° as torn-off corners, as a sum and as a linear pair
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 7 split; the re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes II: the theorem from the triangle sum and the linear pair together

4. Learning Target and Essential Question

Learning target. I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

Essential question. Why do the three angles of every triangle add to the same number?

Building on. T-B1 — supplementary angles and the linear pair; Grade 4 (MA.4.GR.1.3) — angle measures add; Unit 1 — equations in one unknown.

Working toward. Unit 11 — equations and systems; Geometry — the angle sums of polygons and triangle proofs.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 11 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Can a triangle have two right angles? Two obtuse angles? Why not?	DOK 2
Why does an exterior angle equal the two remote interior angles? Use the linear pair.	DOK 3
Each corner of a triangle has two exterior angles. Why are they equal?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Two angles of a triangle are 47° and 68°. Find the third angle.	***65°***	115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180]
2	An isosceles triangle's two base angles are equal. Each is 52°. Find the third angle.	***76°***	128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three]
3	A triangle's remote interior angles are 38° and 77°. Find the exterior angle x.	***115°***	65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]; 295 — the angle around the outside, 360 − 65 [Florida B1G-M, GR.1.5: the reflex angle]
4	A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°?	***B — 105°***	(A) equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Florida B1G-M, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both]
5	An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle.	***75°***	185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Notes II: the exterior angle is not the interior angle beside it]
6	A triangle's angles are x°, $3x^\circ$ and $(2x+12)^\circ$. Find each angle.	***28°, 84° and 68°***	28 alone — reported x, one angle of three [Math Nation names reporting x]
7	An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x?	***C — 5x + 4 = 52 + 3x***	(A) the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]; (B) one remote angle only [the theorem adds both]; (D) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship]
8	A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle.	***73°***	64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum
9	Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain.	***It is the two remote angles added, each more than 0°.***	'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines on the desk: inside, the three make 180; outside, the two far ones added. Shade the two remote angles before writing the equation.

ELL. **Interior**: inside. **Exterior**: outside. **Remote**: far away — not touching. Point to each with the word.

Enrichment. A triangle's angles are in the ratio 2 : 3 : 4. Find each angle, then the exterior angle at the largest one. (40°, 60°, 80°; 100°.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: an exterior angle is the two remote interior angles added, and each of them is more than 0°, so the sum is bigger than either one. A board that says only 'because it is outside' has the claim and not the reason.

Vocabulary. **interior angle**; **exterior angle**; **remote interior angles**; **Triangle Sum Theorem**

Materials. Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 11 test: Find missing angles in triangles — JFJ; Exterior Angle Theorem — FMP; Triangle Angle-Sum Theorem — 6Q6.